

Funnel-Stage SHAP: end-to-end pipeline

DATA

Raw clickstream
42.4M events

Clean + dedup
schema validation

Sessionise
30-min gap

Seeded subsample
485,459 sessions

JOURNEY
RECONSTRUCTION

Stage cut-points
S1 2nd interaction
S2 2nd browsing signal
S3 1st cart

Nested prefixes
 $S1 \subseteq S2 \subseteq S3$
no purchase event

Prefix features per stage
counts · temporal · engagement
entropy · velocity · price

MODELLING

Frozen splits
temporal (headline)
grouped (robustness)

Stage models per S_k
LightGBM · Optuna
5 seeds · calibrated

Sequence model per S_k
GRU on same prefixes

EXPLANATION
AND VALIDATION

Layer 3
faithfulness · stability
seed · cross-paradigm

Layer 1
Interventional TreeSHAP
per stage, grouped

Layer 2
TimeSHAP on the GRU
same prefixes

Validated stage-conditioned explanations
attribution trajectory + quality evidence